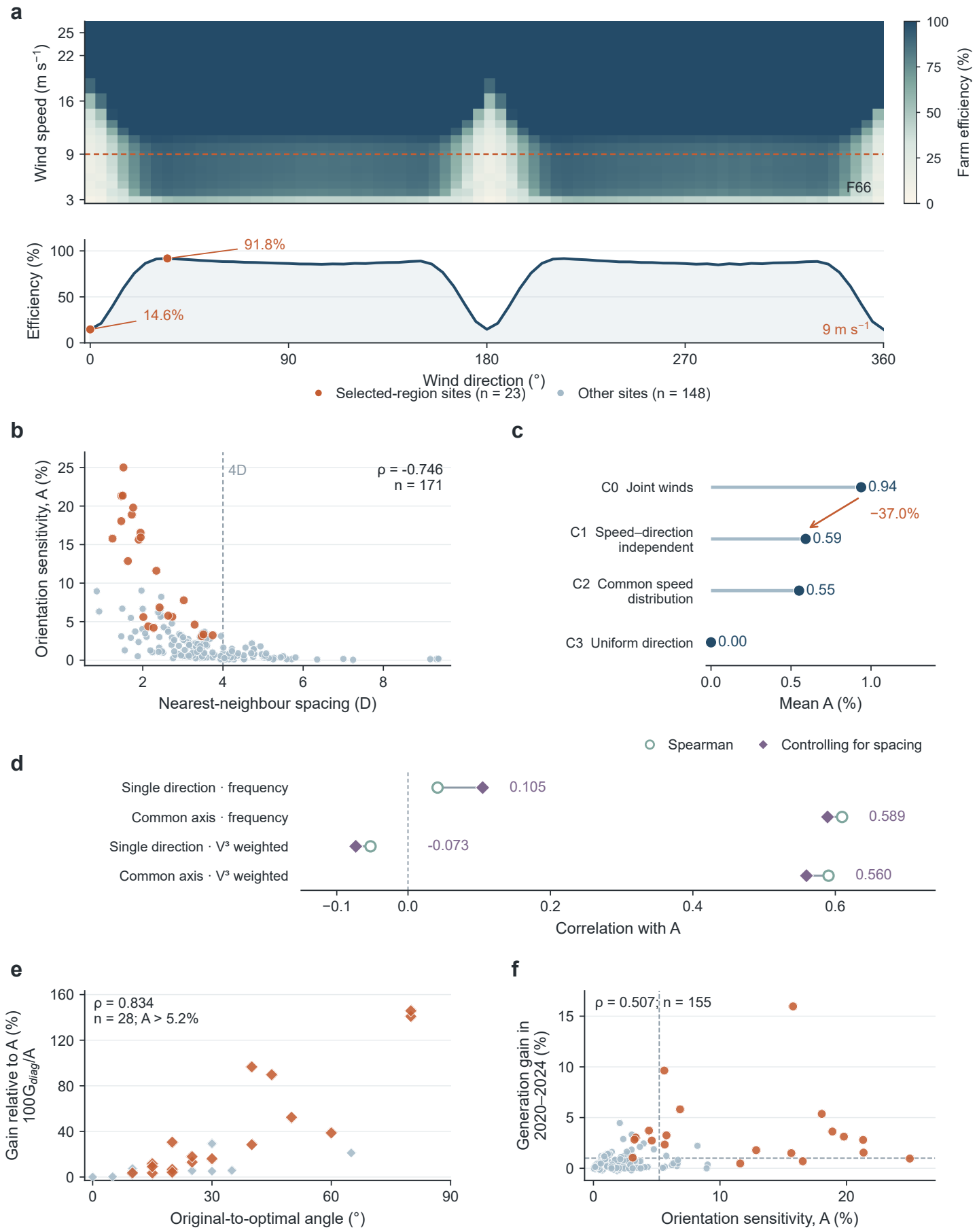

a
Wind speed (m s⁻¹)
25
22
16
9
3
F66
100
75
50
25
0
Farm efficiency (%)
Efficiency (%)
100
50
0
91.8%
14.6%
9 m s⁻¹
0
90
180
270
360
Wind direction (°)
Selected-region sites (n = 23)
Other sites (n = 148)
b
Orientation sensitivity, A (%)
25
20
15
10
5
0
4D
ρ = -0.746
n = 171
2
4
6
8
Nearest-neighbour spacing (D)
c
C0 Joint winds
0.94
−37.0%
C1 Speed–direction independent
0.59
C2 Common speed distribution
0.55
C3 Uniform direction
0.00
0.0
0.5
1.0
Mean A (%)
Spearman
Controlling for spacing
d
Single direction · frequency
0.105
Common axis · frequency
0.589
Single direction · V³ weighted
-0.073
Common axis · V³ weighted
0.560
−0.1
0.0
0.2
0.4
0.6
Correlation with A
e
Gain relative to A (%)
100G_diag/A
160
120
80
40
0
ρ = 0.834
n = 28; A > 5.2%
0
30
60
90
Original-to-optimal angle (°)
f
Generation gain in 2020–2024 (%)
15
10
5
0
ρ = 0.507; n = 155
0
10
20
Orientation sensitivity, A (%)